

Correspondence Theorem for Sets

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Let $p : G \rightarrow Q$ be a surjective map of sets.

Problem 1. Write down natural maps $p_* : \text{Pow}(G) \rightarrow \text{Pow}(Q)$ and $p^* : \text{Pow}(Q) \rightarrow \text{Pow}(G)$ and explain why p_* is surjective, p^* is injective, and p_* is left inverse to p^* .

Solution. $p_* : \text{Pow}(G) \rightarrow \text{Pow}(Q)$ is defined as the following: $\forall X \in \text{Pow}(G), p_*(X) = \{p(a) \mid a \in X\}$. Since p is surjective, for each element of $q \in Q$, $\exists g \in G$ s.t $q(g) = q$. Thus $\forall Y \in \text{Pow}(Q), \forall y \in Y \exists x \in G$ s.t $q(x) = y$. For each $y \in Y$ make a set by taking 1 $x \in G$ s.t $p(x) = y$. By definition of $p_*(X) = Y$. Thus there exists a $X \in \text{Pow}(G)$ s.t $p_*(X) = Y$. Thus p_* is surjective.

$p^* : \text{Pow}(Q) \rightarrow \text{Pow}(G)$ is defined as the following: $\forall Y \in \text{Pow}(Q), p^*(Y) = \{x \mid x \in G, p(x) \in Y\}$. Suppose $A, B \in \text{Pow}(G)$ and $p^*(A) = p^*(B)$ thus means $\forall x \in p^*(A) = p^*(B), p(x) \in A \cap B$. Furthermore since p is surjective by the definition of p^* , $\nexists a \in A$ s.t $\nexists x \in p^*(A) p(x) \neq a$. This is the same for B . Furthermore by definition of a function, $x \in G$ maps 1 and only 1 element in G . Since there are no other values in A or B except the values in $A \cap B$ then $A = B$. Thus p^* is injective.

$\forall B \in \text{Pow}(G), p_*(p^*(B)) = p_*(\{x \mid x \in G, p(x) \in B\})$ which is equal to all elements $\{p(x) \mid x \in p^*(B)\}$. However since p is surjective function each element in Q has at least 1 element assigned to it and each element in G is assigned to 1 element. Thus $\forall x \in p^*(B)$, it means $p(x) \in B$ by definition of the preimage. Thus $p_*(p^*(B)) \subseteq B$. $\forall b \in B$, there exists a x s.t $p(x) = b$ by definition of surjective function since p is surjective and by definition of a preimage $x \in p^*(B)$ and $p(x) = b \in p_*(p^*(B))$ by definition of image. Thus $B \subseteq p_*(p^*(B))$, thus $p_*(p^*(B)) = B$. Thus p_* is left inverse to p^* . ■

Problem 2a. Explain why using 1 its reasonable to view $\text{Pow}(Q)$ as a subset of $\text{Pow}(G)$. Then its natural to ask, which subset?

Solution. Since there is a natural injective map from $\text{Pow}(Q) \rightarrow \text{Pow}(G)$ just use the injective map as a intermediary. $\text{Pow}(Q)$ are the subsets $S = \{p^*(Y) \mid Y \in \text{Pow}(Q)\} \subset \text{Pow}(G)$. p^* is injective and p^* is surjective in S because the image of p^* is S . Thus $\text{Pow}(Q)$ is in bijection with S ■

Claim 2b. Z is "closed under the equivalence relation" where $a \sim b \iff p(a) = p(b)$, and that this holds iff $Z = p^*(p_*(Z))$.

Proof. ($\implies$): Z is "closed under the equivalence relation" where $a \sim b \iff p(a) = p(b)$. Thus $\forall a \in Z \forall b \in G$ s.t $p(a) = p(b)$ then $b \in Z$. In otherwords, $\forall a \in Z, p^*(p_*(\{a\})) \in Z$. Thus $Z = p^*(p_*(Z))$

($\impliedby$): Suppose $Z = p^*(p_*(Z))$. Then $\forall a \in Z, p^*(p_*(\{a\})) \in Z$. Thus if $\forall a \in Z, \forall b \in G$ s.t $p(a) = p(b)$ then $b \in Z$. Thus by definition of the equivalence relation $\forall a \in Z$, if $a \sim b \implies b \in Z$. Thus Z is closed under the equivalence relation. □

Claim 2c. Show Z in $\text{Pow}(G)$ lies in $\text{Pow}(Q)$ iff Z is "closed under the equivalence relation" where $a \sim b \iff p(a) = p(b)$.

Proof. ($\implies$) Suppose $Z \in \text{Pow}(G)$ and $Z \in \text{Pow}(Q)$. Then $Z = p_*(Y)$ of some $Y \in \text{Pow}(Q)$, by the natural injective map of the preimage which implies that $Z \in \text{Pow}(G) \cap \text{Pow}(Q)$. Thus $Z = \{x \mid p(x) \in Y\}$. $p_*(p^*(Z)) = p_*(p^*(p_*(Y))) = p_*(Y) = Z$ because p^* is the left inverse of p_* by Problem 1 thus $Z = p_*(p^*(Z))$. By the previous proof, then Z is "closed under the equivalence relation" where $a \sim b \iff p(a) = p(b)$.

($\impliedby$): Suppose $Z \in \text{Pow}(G)$ is "closed under the equivalence relation" where $a \sim b \iff p(a) = p(b)$. Let $Y := p_*(Z)$, $Y \in \text{Pow}(Q)$. Then by Claim 2b, $Z = p^*(p_*(Z)) = p^*(Y)$. Thus by the natural injective map since $Z = p^*(Y)$ thus Z is in bijection with Y , since $Y \in \text{Pow}(Q)$ and then $Z \in \text{Pow}(Q)$ $\square$

Problem 3. Let W be another set, explain why $\text{Hom}(Q, W)$ is naturally a subset of $\text{Hom}(G, W)$. Which subset?

Solution. $Q \cong G/\sim$ under the previously defined equivalence relation because Q is a surjective function. Thus $\text{Hom}(Q, W) \cong \text{Hom}(G/\sim, W)$ and is the subset of functions in $\text{Hom}(G, W)$ which factors through p . Let $\phi : \text{Hom}(G/\sim, W) \rightarrow \text{Hom}(G, W)$ which takes $F \rightarrow F \circ p$. Suppose $\forall F, G \in \text{Hom}(Q, W)$, that $\phi(F) = \phi(G)$ thus $F \circ p = G \circ p$. Thus $\forall x \in G$, $F(p(x)) = G(p(x))$ and $G(p(x)) = G([x])$ thus $F([x]) = G([x])$ for all $x \in G$ and thus all $[x] \in G/\sim$ thus $F = G$. Thus ϕ is injective. Furthermore by definition of ϕ , the subset of $\text{Hom}(G, W)$ which $\text{Hom}(G/\sim, W) \cong \text{Hom}(Q, W)$ is the subset of functions which factor through p . $\blacksquare$

Problem 4. Now suppose G and Q are groups, and p is a (surjective) homomorphism. Explain why p_* and p^* restrict to maps between subgroups and deduce the correspondence theorem for groups from the above.

Proof. Because p is a homomorphism, it preserves group structure by definition and takes identities to identities and inverses to inverses. Let $F \leq G$ be any subgroup. Then by definition of group $e \in F$ and by definition of a homomorphism $p(e) = e \in p_*(F)$. Furthermore, for all $g \in F$ $g^{-1} \in F$ by definition of a group and $p(g) \in p(F)$ by definition of image. By definition of a homomorphism $p(g)^{-1} = p(g^{-1}) \in p_*(F)$. Finally $\forall a, b \in p_*(F)$ by definition of image there exists $x, y \in F$ s.t $p(x) = a$ and $p(y) = b$. $ab = p(x)p(y) = p(xy)$ by definition of a homomorphism and $xy \in F$ thus $p(xy) = ab \in p_*(F)$. Thus $p_*(F)$ is a subgroup. Thus p_* takes subgroups to subgroups.

Let $R \leq Q$ be a subgroup of Q . Then $e_Q \in R$ by definition of a subgroup and by definition of a homomorphism $p(e_G) = e_Q$ then $e_G \in p^*(R)$ by definition of a preimage. $\forall r \in R$, $r^{-1} \in R$ by definition of a subgroup. Since p is surjective, there exists a $x \in G$ s.t $p(x) = r$ by definition of a preimage $x \in p^*(R)$. $p(x^{-1}) = p(x)^{-1} = r^{-1} \in R$ thus $x^{-1} \in p^*(R)$. Thus every element in $p^*(R)$ has inverses. $\forall a, b \in p^*(R)$, by definition of a preimage $p(a), p(b) \in R$. $p(ab) = p(a)p(b) \in R$ because R is subgroup and by definition of preimage since $p(ab) = p(a)p(b) \in R$, then $ab \in p^*(R)$ thus $p^*(R)$ is closed. Thus $p^*(R)$ is closed.

By problem 2, we know that elements of the $\text{Pow}(Q)$ are in bijection with elements of $\text{Pow}(G)$ which are closed under the equivalence relation such that $a \sim b \implies p(a) = p(b)$. Moreover in the contexts of groups since p is a surjective homomorphism and p_* and p^* can be restricted maps between subgroups, then the subgroups of Q are in bijection with the subgroups of G such that are closed under

the equivalence relation. By 2b, for a subgroup $Z \leq G$ which is in bijection with the subgroup of Q , $Z = p_*(p^*(Z))$. Thus since $e_G \in p^*(Z)$, then by definition $p_*(p^*(Z))$ should contain all elements $a \in G$ s.t $p(a) = e$, thus $\ker p \subset p_*(p^*(Z))$. Thus the subgroups of Q are in bijection with the subgroups of G which contain the kernel p . $\square$